Class 22: Confinement and Quantization: Part 1

In the last class we noted that when an electron travels through a solid, it is not reasonable to assume that there is no interaction with the solid, or that the interaction is featureless.

We identified an approximation of the potential landscape that the electron experiences in the solid.

Just to get an idea of the numbers involved, let us assume that we have solid that is made up entirely using atoms of an element with atomic number 50. Further, let us assume that each of the atoms in the solid releases one electron to the free electron gas. In terms of the approximate potential landscape we have just identified, this means that 49 out of the 50 electrons associated with each ion, are confined to the nearly 1 Å space adjacent to the location of the ionic core, or are 'bound' to the location of the ionic core. The remaining one electron per ion is free to roam the solid but is confined to the physical extent of the solid. The extent of the solid can be of the order of several meters. Therefore, while all of the electrons within the solid are confined, the bound electrons are confined to a very small region, which is ten orders of magnitude smaller than the region over which the nearly free electrons are confined.

In the calculations so far we have neglected the 49 electrons that are bound to the ionic cores, and only focused our attention to the one electron per ionic core that has been released to be part of the free electron gas. We have adopted this approach, since the properties we are examining are affected only by the nearly free electrons and not the bound electrons.

As indicated in our discussion of the Drude-Sommerfeld model, when the electrons fill up the states available to the nearly free electrons, starting from the lowest available energy level, the highest energy level that gets filled at 0 Kelvin, is called the Fermi energy level. In terms of the energy levels shown in Figure 22.1 below, when an electron is at infinity, it has zero potential energy due to the solid. As it gets closer to the solid, the energy becomes increasingly negative, in a relative sense. The nearly free electrons exist in the energy window between zero potential and a somewhat negative potential. In this potential window, the nearly free electrons fill the lowest energy level available first, and then progressively fill higher and higher energy levels, till the system runs out of free electrons. The highest energy level occupied, within the levels available to the nearly free electrons, is then the Fermi energy level.

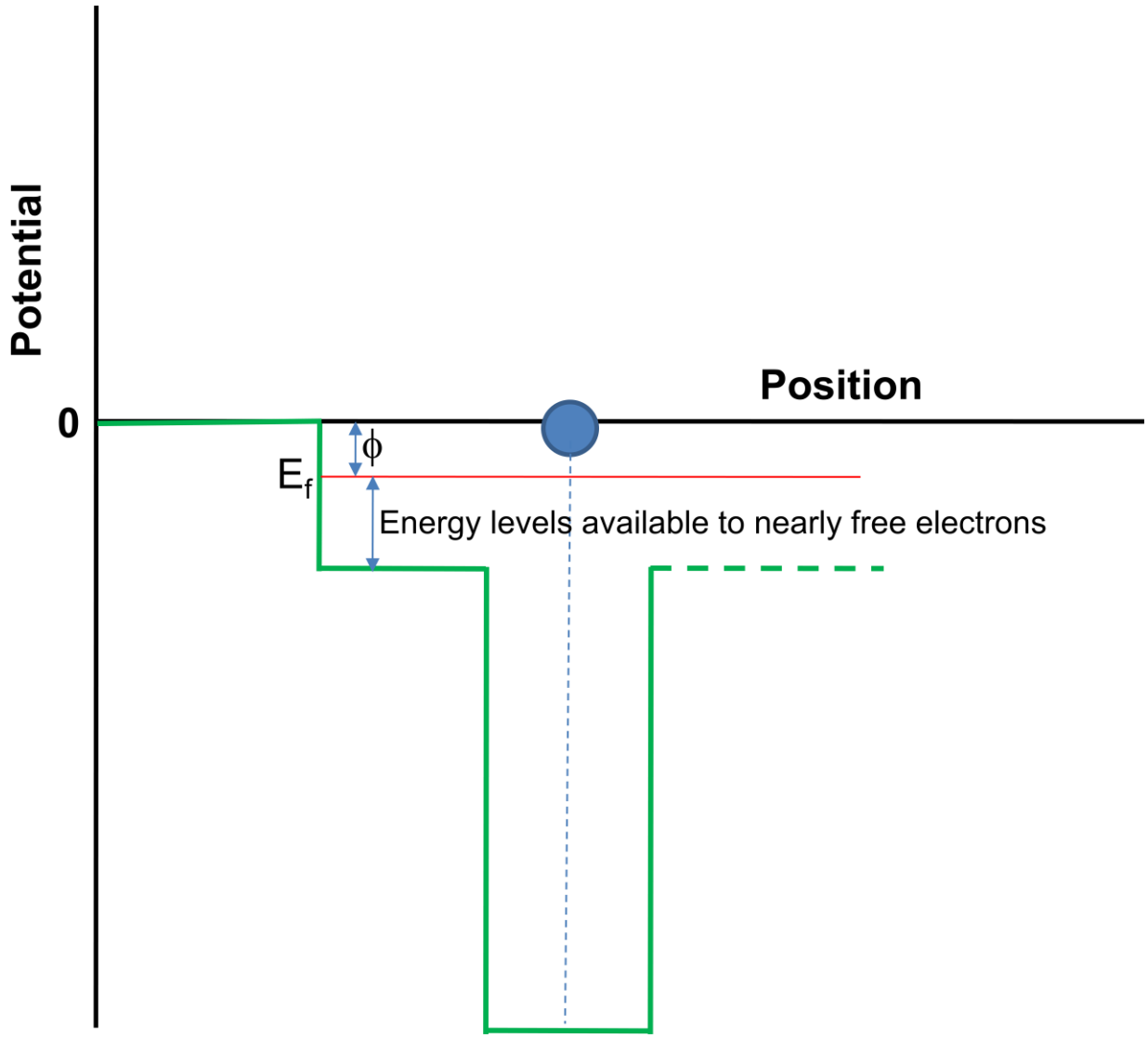


Figure 22.1: The energy levels available to the nearly free electrons and the Fermi energy level E_f, as well as work function ϕ, in a metallic sample.

When external energy is supplied to the sample in the form of light, and a photo electron is ejected from the sample, the least amount of energy required will be that required to pull the highest energy electron available in the sample, and take it to infinity with respect to the sample. The highest energy electrons available in the metal are the electrons at the Fermi energy. The energy that will therefore be required to remove this electron, will be the difference between the Fermi energy and the zero potential corresponding to the escaped electron. This energy required, is defined as the work function and is also indicated in Figure 22.1 above.

Based on the above discussion, we now have an understanding of the confinement experienced by the electrons in a solid and the relationships between quantities such as the Fermi energy and

the work function of a solid. Let us now use this picture to understand behavior of electrons in a solid.

In general, here are two types of electrons:

1) Truly free electrons
2) Confined Electrons – which can either be the nearly free electrons, or the bound electrons.

To understand how electrons behave, we have to employ quantum mechanical tools to the problem. However, we will initially use a much simpler approach, which will give us an idea of the key concepts involved and will enable us to understand the quantum mechanical approach and its results more easily. To initiate the simpler approach, let us look at how the quantum mechanical approach will proceed and then make the simplification at the appropriate stage.

Quantum mechanical description acknowledges that there exists a wave particle duality. For the moment, let us look at electrons as waves.

Based on the de Broglie equation, associated with a particle having a momentum p, there exists a wavelength λ such that:

$$p\lambda = h$$

Where h is the Planck's constant. Rearranging, we have:

$$p = \frac{h}{\lambda}$$

Defining $\hbar = \frac{h}{2\pi}$, and a vector called the wave vector $k = \frac{2\pi}{\lambda}$, we can write the equation for the momentum as follows:

$$p = \frac{h}{2\pi} \times \frac{2\pi}{\lambda} = \hbar k$$

Since energy and momentum are related as:

$$E = \frac{p^2}{2m}$$

We have:

$$E = \frac{\hbar^2 k^2}{2m}$$

Therefore the energy, momentum, and wave vector corresponding to a particle, are related. Any restriction on one of these quantities, will automatically restrict the other two quantities as well.

We have previously assumed that electrons in a solid have specific allowed, or discrete energy levels ε_1, ε_2, ε_3, …It is now time to examine why the energy levels of electrons in a solid cannot be continuous, and what these discrete energy values are.

To understand what energy values and corresponding wavelengths are allowed for electrons in a solid, let us begin by examining a simpler analogy.

Consider a string, that is tied to a wall on one end and with the other end free, as shown in Figure 22.2 below. The question we wish to ask is: what are the waves that can be supported by this string?

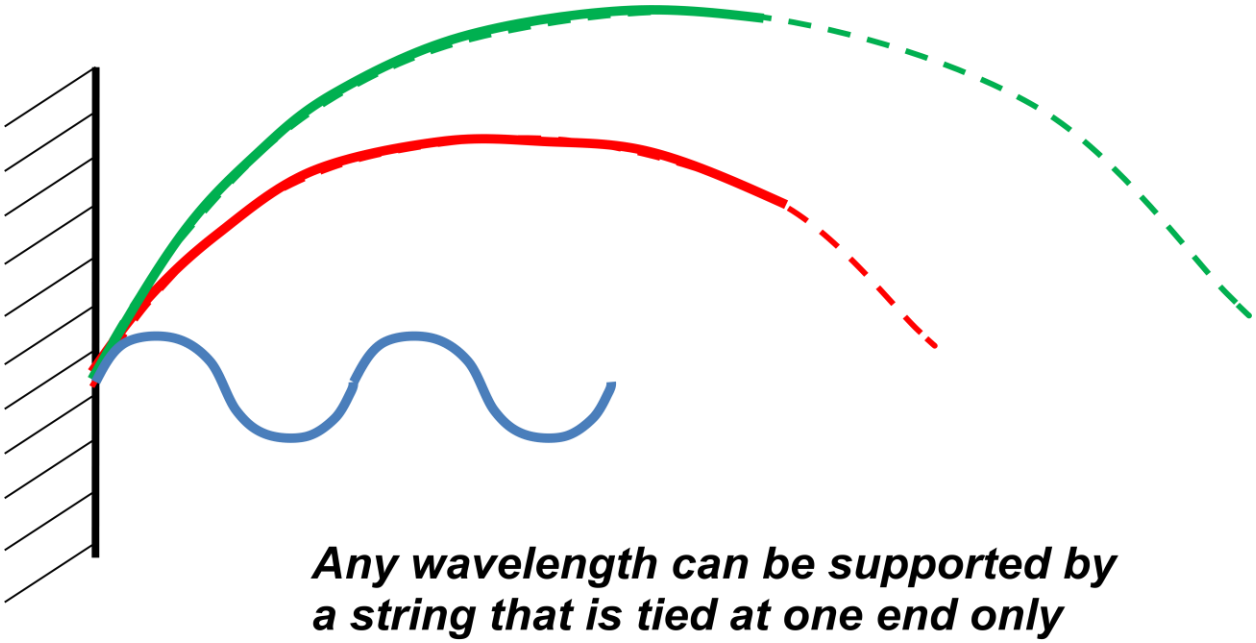
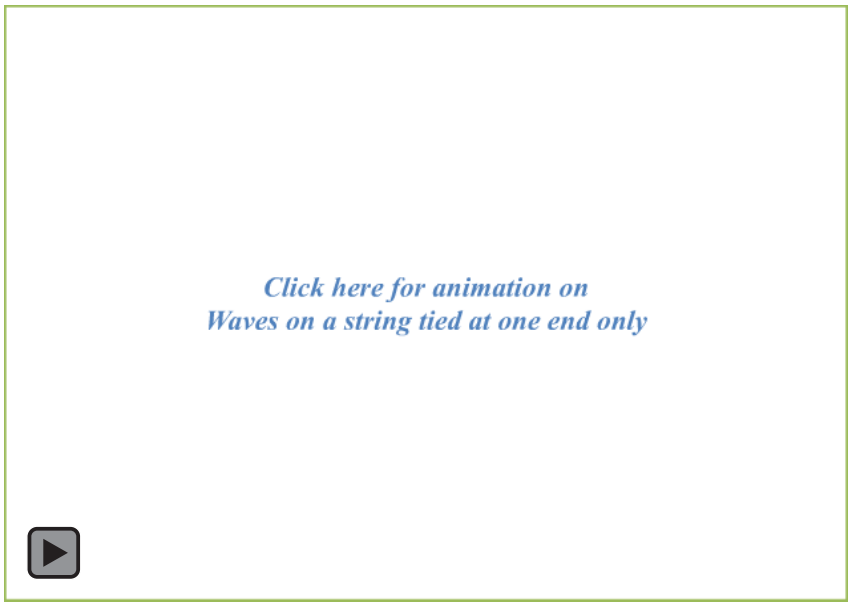


Figure 22.2: Some examples of waves that can be supported by string tied at only one end.

Since one end is free, any and all wavelengths can be supported by the string – or in other words, the string can adopt a shape consistent with any λ.



Animation of figure 22.2

Now consider a string that is tied at both ends as shown in Figure 22.3 below.

Only specific wavelengths can be supported by a string tied at both ends, since the ends have to be nodes.

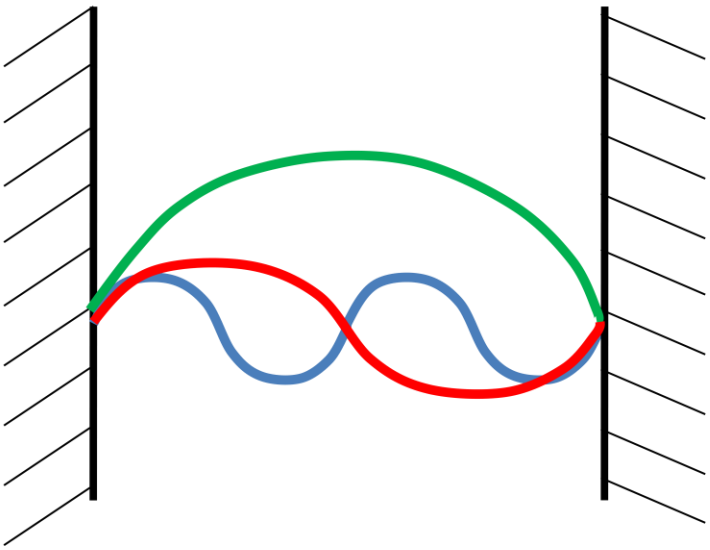


Figure 22.3: Some examples of waves that can be supported by a string tied at both ends.

Click here for animation on
Waves on a string tied at both ends

Animation of figure 22.3

The significance of the fact that both ends of the string are tied is that at those two ends the string can only support zero displacement. In other words, the waves supported by the string must have nodes at those two end points, or the waves must be such that the distance between the ends of the string must correspond to integral multiples of half wavelengths.

If the distance between the walls where the string is tied is a, then the wavelengths supported by the string can only be of the kind that satisfies one of the following relationships:

$$a = \frac{\lambda}{2}$$

Or

$$a = \frac{2\lambda}{2}$$

Or

$$a = \frac{3\lambda}{2}$$

etc. The general form being $a = \frac{n\lambda}{2}$ where n is an integer. These are the only values of wavelength that are allowed for a string tied at two ends.

Therefore for a string that is free or tied at only one end, there are no restrictions on the wavelengths that can be supported. Therefore values of energy corresponding to these wavelengths can also be supported by the string. The continuous values of allowed wavelengths permits continuous values of energy to be supported by the string as well.

For a string tied at both ends, not all values of wavelength can be supported due to the constraint that the ends of the string must be nodes. Therefore only specific values of energy corresponding to these specific values of wavelength, can be supported by the string. For every discrete value of λ permitted (or k permitted), there is a corresponding discrete value of energy, which is permitted. When $a = \frac{\lambda}{2}$, a corresponding energy value is permitted to be displayed by the string. The next nearest value of energy that can be displayed by the string will correspond to the situation where $a = \frac{2\lambda}{2}$. Between these two energy values, no other energy value can be displayed by the string. By confining the string at both ends, we have prevented it from supporting a continuous set of energy values. This discretization of energy values is called ‘Quantization’. In other words, ‘confinement’ of the string (tying both ends) has directly resulted in ‘quantization’ (discretization) of allowed energy levels.

In the next class we will look at the quantum mechanical approach to handling this problem of confinement, and examine if the results obtained through the use of the analogy of waves on a string is consistent with the results obtained from quantum mechanics.